

```

struct BIT {
    int n;
    vector<ll> bit;
    int lowbit(int x) { return x & -x; }
    BIT(int _n) : n(_n + 1) { bit = vector<ll>(_n + 1, 0); }
    void update(int x, ll v) { // 將 x 的值加上 v
        for (; x < n; x += lowbit(x)) bit[x] += v;
    }
    ll query(int x) { // 查詢區間 [1, x] 的總和
        ll ret = 0;
        for (; x; x -= lowbit(x)) ret += bit[x];
        return ret;
    }
    ll query(int l, int r) { // 查詢區間 [l, r] 的總和
        return query(r) - query(l - 1);
    }
    int kth(int k) {
        int x = 0;
        for (int i = 1 << __lg(n); i; i >>= 1) {
            if (x + i <= n && k >= bit[x + i - 1]) {
                x += i, k -= bit[x - 1];
            }
        }
        return x;
    }
};

```

2.3 二維 BIT

```
struct BIT {
    int n, bit[MXN][MXN];
    int lowbit(int x) { return x & -x; }
    BIT(int _n) : n(_n + 1) { memset(bit, 0, sizeof(bit)); }
    void update(int x, int y, int val) { // 更新 (x, y)
        for (; x < n; x += lowbit(x))
            for (; y < n; y += lowbit(y)) bit[x][y] += val;
    }
    int query(int x, int y) { // 查詢 (1, 1) ~ (x, y) 的和
        int ans = 0;
        for (; x; x -= lowbit(x))
            for (; y; y -= lowbit(y)) ans += bit[x][y];
        return ans;
    }
    int range_query(int a, int b, int x, int y) {
        return query(x, y) - query(x, b - 1) -
            query(a - 1, y) + query(a - 1, b - 1);
    }
} bit;
```

2.4 Segment Tree

```
#define cl (i << 1)
#define cr (i << 1 | 1)
#define NO_TAG 0
struct SegmentTree { // 1-base
    int n;
    vector<int> seg, tag;
    SegmentTree(int _n) { // 空的 SegmentTree
        n = _n;
        seg.resize(n * 4);
        tag.resize(n * 4);
    }
    void push(int i, int l, int r) {
        if (tag[i] != NO_TAG) {
            seg[i] += tag[i]; // update by tag
            if (l != r) {
                tag[cl] += tag[i]; // push
                tag[cr] += tag[i]; // push
            }
            tag[i] = 0;
        }
    }
    void pull(int i, int l, int r) {
        int mid = (l + r) >> 1;
        push(cl, l, mid);
        push(cr, mid + 1, r);
        seg[i] = max(seg[cl], seg[cr]); // pull (操作改這)
    }
    void build(int i, int l, int r) {
        if (l == r) {
            seg[i] = arr[l]; // 初始值
            return;
        }
        int mid = (l + r) >> 1;
        build(cl, l, mid);
        build(cr, mid + 1, r);
        pull(i, l, r);
    }
    void update(int i, int l, int r, int ql, int qr,
                int v) {
        push(i, l, r);
        if (ql <= l && r <= qr) {
            tag[i] += v;
            return;
        }
        int mid = (l + r) >> 1;
        if (ql <= mid) update(cl, l, mid, ql, qr, v);
        if (qr > mid) update(cr, mid + 1, r, ql, qr, v);
        pull(i, l, r);
    }
    int query(int i, int l, int r, int ql, int qr) {
        push(i, l, r);
        if (ql <= l && r <= qr) {
            return seg[i];
        }
        int mid = (l + r) >> 1, ans = -INF;
        if (ql <= mid) // (操作改這)
            ans = max(ans, query(cl, l, mid, ql, qr));
        if (qr > mid) // (操作改這)
            ans = max(ans, query(cr, mid + 1, r, ql, qr));
    }
};
```

```
return ans;
}
// 區間 [ql, qr] 加值 v
void update(int ql, int qr, int v) {
    update(1, 1, n, ql, qr, v);
}
// 查詢區間 [ql, qr]
int query(int ql, int qr) {
    return query(1, 1, n, ql, qr);
}
};
```

2.5 Treap

```
struct Treap {
    int sz, val, pri, tag;
    Treap *l, *r;
    Treap(int _val) : val(_val) {
        sz = 1;
        pri = rand(); // 隨機產生優先度
        tag = 0;
        l = r = NULL;
    }
};
int size(Treap *a) { return a ? a->sz : 0; }
void pull(Treap *a) {
    a->sz = size(a->l) + size(a->r) + 1;
}
Treap *merge(Treap *a, Treap *b) {
    // val of a is always bigger than val of b
    if (!a || !b) return a ? a : b;
    if (a->pri > b->pri) {
        a->r = merge(a->r, b);
        pull(a);
        return a;
    } else {
        b->l = merge(a, b->l);
        pull(b);
        return b;
    }
}
void split(Treap *t, int k, Treap *&a, Treap *&b) {
    // a < k, b >= k
    if (!t) {
        a = b = NULL;
        return;
    }
    if (k <= t->val) {
        b = t;
        split(t->l, k, a, b->l);
        pull(b);
    } else {
        a = t;
        split(t->r, k, a->r, b);
        pull(a);
    }
}
Treap *add(Treap *t, int v) {
    Treap *val = new Treap(v);
    Treap *l = NULL, *r = NULL;
    split(t, v, l, r);
    return merge(merge(l, val), r);
}
Treap *del(Treap *t, int v) {
    Treap *l, *mid, *r, *temp;
    split(t, v, l, temp);
    split(temp, v + 1, mid, r);
    return merge(l, r);
}
int position(Treap *t, int p) { // base 1
    if (size(t->l) + 1 == p) return t->val;
    if (size(t->l) < p)
        return position(t->r, p - size(t->l) - 1);
    else return position(t->l, p);
}
int query(Treap *t, int k) { // num of >= k
    if (!t) return 0;
    if (t->val == k) return size(t->l) + 1;
    if (t->val > k) return query(t->l, k);
    return size(t->l) + 1 + query(t->r, k);
}
};
```

2.6 平板電腦

```
#include <bits/extc++.h>

#include <ext/pb_ds/assoc_container.hpp>
#include <ext/rope>
#include <functional>
#include <queue>
using namespace std;
using namespace __gnu_cxx;
using namespace __gnu_pbds;
typedef tree<int, null_type, less<int>, rb_tree_tag,
            tree_order_statistics_node_update>
    set_t;
typedef cc_hash_table<int, int> umap_t;
typedef priority_queue<int> heap;
// Insert some entries into s.
set_t s;
s.insert(12);
s.insert(505);
// The order of the keys should be: 12, 505.
assert(*s.find_by_order(0) == 12);
assert(*s.find_by_order(3) == 505);
// The order of the keys should be: 12, 505.
assert(s.order_of_key(12) == 0);
assert(s.order_of_key(505) == 1);
// Erase an entry.
s.erase(12);
// The order of the keys should be: 505.
assert(*s.find_by_order(0) == 505);
// The order of the keys should be: 505.
assert(s.order_of_key(505) == 0);
heap h1, h2;
h1.join(h2);
rope<char> r[2];
r[1] = r[0]; // persistenet
string t = "abc";
r[1].insert(0, t.c_str());
r[1].erase(1, 1);
cout << r[1].substr(0, 2);
```

3 Mathematics

3.1 公式們

$$\sum_{k=1}^n n = \frac{n(n+1)}{2}, \quad \sum_{k=1}^n n^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{k=1}^n n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

$$\sum_{k=1}^n n^4 = n(1+n)(1+2n)(-1+3n+3n^2)/30$$

$$\sum_{k=1}^n n^5 = n^2(n+1)^2(2n^2+2n-1)/12$$

錯位排列

$$DP_i = \begin{cases} 0 \vee 1, & i = 1 \vee 2 \\ (i-1)(DP_{i-1} \times DP_{i-2}), & i \geq 3 \\ iDP_{i-1} + (-1)^i & \text{另一種關係式} \end{cases}$$

$$P_k^n = \frac{n!}{k!(n-k)!}, \quad C_k^n = \frac{n!}{(n-k)!}, \quad H_k^n = C_{n-1}^{n+k-1}$$

$$\pi = \arccos(-1), \quad e^{i\theta} = \cos \theta + i \sin \theta, \quad pV = nTR$$

$$a^{\varphi(m)} \equiv 1 \pmod m$$

$$a^b \equiv a^{b \bmod \varphi(m)-1} \pmod m$$

$$\varphi(m) = m - 1, \text{ When } m \text{ is Prime}$$

3.2 階乘 & 模逆元

```
ll fac[MXN], inv[MXN];
void init() {
    fac[0] = 1; // 0! = 1
    for (ll i = 1; i < MXN; i++)
        fac[i] = fac[i - 1] * i % MOD; // n! = n * (n-1)!
    inv[MXN - 1] = power(fac[MXN - 1], MOD - 2);
    for (ll i = MXN - 2; i >= 0; i--)
        inv[i] = inv[i + 1] * (i + 1) % MOD; // (n!)^(-1)
}
```

3.3 GCD and LCM

```
ll gcd(ll a, ll b) {
    if (b && a) // 輾轉先除法 -> 交叉取餘數直到不能再取
        while ((a %= b) && (b %= a)) {}
    return a + b;
}

ll lcm(ll a, ll b) { return a * b / gcd(a, b); }
```

3.4 EXGCD

$$ax + by = gcd(a, b)$$

```
int exgcd(int a, int b, ll &x, ll &y) {
    if (b == 0) {
        x = 1, y = 0;
        return a;
    }
    int now = exgcd(b, a % b, y, x);
    y -= a / b * x;
    return now;
}
```

3.5 Fast Power

```
ll power(ll x, ll y) {
    ll ret = 1;
    while (y) {
        if (y & 1) (ret *= x) %= MOD;
        (x *= x) %= MOD;
        y >>= 1;
    }
    return ret;
}
```

3.6 矩陣乘法

```
struct mat { // mat t(r,c);
    ll a[200][200], r, c;
    mat(int _r, int _c) : r(_r), c(_c) {
        memset(a, 0, sizeof(a));
    }
    void build() { // 單位矩陣
        for (int i = 0; i < r; ++i) a[i][i] = 1;
    }
};
mat operator*(mat x, mat y) {
    mat z(x.r, y.c);
    for (int i = 0; i < x.r; ++i)
        for (int j = 0; j < x.c; ++j)
            for (int k = 0; k < y.c; ++k)
                (z.a[i][j] += x.a[i][k] * y.a[k][j] % MOD) %= MOD;
    return z;
}
```

3.7 FFT

使用前要先跑過 pre_fft(); n 必須是 2^k。

```
const int MAXN = 262144 << 1; // (must be 2^k)
// before any usage, run pre_fft() first
typedef long double ld;
typedef complex<ld> cplx; // real(), imag()
const ld PI = acosl(-1);
const cplx I(0, 1);
cplx omega[MAXN + 1];
void pre_fft() {
    for (int i = 0; i <= MAXN; i++)
        omega[i] = exp(i * 2 * PI / MAXN * I);
}
void fft(int n, cplx a[], bool inv = false) {
    int basic = MAXN / n;
    int theta = basic;
    for (int m = n; m >= 2; m >>= 1) {
        int mh = m >> 1;
        for (int i = 0; i < mh; i++) {
            cplx w = omega[inv ? MAXN - (i * theta % MAXN)
                               : i * theta % MAXN];
            for (int j = i; j < n; j += m) {
                int k = j + mh;
                cplx x = a[j] - a[k];
                a[j] += a[k];
                a[k] = w * x;
            }
        }
        theta = (theta * 2) % MAXN;
    }
    int i = 0;
    for (int j = 1; j < n - 1; j++) {
        for (int k = n >> 1; k > (i ^= k); k >>= 1) {}
        if (j < i) swap(a[i], a[j]);
    }
    if (inv)
        for (i = 0; i < n; i++)
            a[i] /= n;
```

```

}
cplx arr[MAXN + 1];
inline void mul(int _n, ll a[], int _m, ll b[],
               ll ans[]) {
    int n = 1, sum = _n + _m - 1;
    while (n < sum)
        n <= 1;
    for (int i = 0; i < n; i++) {
        double x = (i < _n ? a[i] : 0),
               y = (i < _m ? b[i] : 0);
        arr[i] = complex<double>(x + y, x - y);
    }
    fft(n, arr);
    for (int i = 0; i < n; i++)
        arr[i] = arr[i] * arr[i];
    fft(n, arr, true);
    for (int i = 0; i < sum; i++)
        ans[i] = (long long int)(arr[i].real() / 4 + 0.5);
}

```

3.8 質數表

```

// isprime, 最大質因數, 質因數數量
int isPrime[MXN], fac[MXN], num[MXN];
void getPrimes() {
    isPrime[1] = 1, fac[1] = num[1];
    for (int i = 2; i < MXN; ++i) {
        if (isPrime[i]) continue;
        for (int j = i; j < MXN; j += i) {
            if (i != j) isPrime[j] = 1;
            fac[j] = i;
            num[j]++;
        }
    }
}
vector<int> ret; // 質因數分解
void div(int x) {
    for (; x > 1; x /= fac[x])
        ret.push_back(fac[x]);
}

```

3.9 歐拉函數

```

φ(x) = 所有小於等於 x 且和 x 互質的數的數量

int phi(int n) { // O(sqrtN)
    int res = n, a = n;
    for (int i = 2; i * i <= a; i++) {
        if (a % i == 0) {
            res = res / i * (i - 1);
            while (a % i == 0)
                a /= i;
        }
    }
    if (a > 1) res = res / a * (a - 1);
    return res;
}

int phi[MXN]; // 建表 最大1e7
void phi_table(int n) {
    phi[1] = 1;
    for (int i = 2; i <= n; ++i) {
        if (phi[i]) continue;
        for (int j = i; j <= n; j += i) {
            if (phi[j] == 0) phi[j] = j;
            phi[j] = phi[j] / i * (i - 1);
        }
    }
}

```

3.10 Miller-Rabin Algorithm

$$\text{Magic} = \begin{cases} \{2, 7, 61\}, & n < 4,759,123,141 \\ \{2, 13, 23, 1662803\}, & n < 1,122,004,669,633 \\ \{2, 3, 5, 7, 11, 13\}, & n < 3,474,749,660,383 \\ \{2, 325, 9375, 28178, \\ 450775, 9780504, 1795265022\} & n < 2^{64} \end{cases}$$

```

ll mul(ll x, ll y, ll mod) {
    ll ret = x * y - (ll)((long double)x / mod * y) * mod;
    return ret < 0 ? ret + mod : ret;
    // return x * y % mod; // for __int128
}

```

```

ll magic[] = {}; // 這邊填入要用於測試的數列
ll S = 3; // 測試的數列數字的數量
bool witness(ll a, ll n, ll u, int t) {
    if (!a) return 0;
    ll x = power(a, u, n);
    for (int i = 0; i < t; i++) {
        ll nx = mul(x, x, n);
        if (nx == 1 && x != 1 && x != n - 1) return 1;
        x = nx;
    }
    return x != 1;
}
bool miller_rabin(ll n) {
    int s = S; // magic number size
    // iterate s times of witness on n
    if (n < 2) return false;
    if (!(n & 1)) return (n == 2);

    // 將 n - 1 寫成 u * (2 ^ t) 的形式
    ll u = n - 1;
    int t = 0;
    while (!(u & 1))
        u >>= 1, t++;

    while (s--) {
        ll a = magic[s] % n;
        if (witness(a, n, u, t)) return false;
    }
    return true;
}

```

3.11 Pollard's Rho Algorithm

```

vector<ll> ret;
ll mul(ll x, ll y, ll mod) {
    ll ret = x * y - (ll)((long double)x / mod * y) * mod;
    return ret < 0 ? ret + mod : ret;
    // return x * y % mod; // for __int128
}
ll f(ll x, ll c, ll mod) {
    return (mul(x, x, mod) + c) % mod;
}
ll pollard_rho(ll n) {
    ll c = 1, x = 0, y = 0, p = 2, q, t = 0;
    while (t++ % 128 or gcd(p, n) == 1) {
        if (x == y) c++, y = f(x = 2, c, n);
        if (q = mul(p, abs(x - y), n)) p = q;
        x = f(x, c, n);
        y = f(f(y, c, n), c, n);
    }
    return gcd(p, n);
}
void fact(ll x) { // 透過遞迴找質數
    if (miller_rabin(x)) {
        ret.push_back(x);
        return;
    }
    ll f = pollard_rho(x);
    fact(f);
    fact(x / f);
}

```

3.12 高斯消去法

```

/**
 * mat[n][m] 為 n 個 m 元一次連列方程式的增廣矩陣
 * ans[m] 輸出 x_m 的解
 * led[n] 紀錄第 n 列的領導係數的位置
 * 若無解則回傳 false
 */
ll mat[MAX_N][MAX_N], ans[MAX_N], led[MAX_N];
bool gaussian(int n, int m) {
    for (int i = 0; i < n; i++) { // 高斯消元法
        // 第 i 列的領導項
        while (led[i] <= m && mat[i][led[i]] == 0) led[i]++;

        // 當每項係數皆為 0 時, mat_{im} == b_i != 0 => 無解
        if (led[i] == m && mat[i][m] != 0) return false;

        int c = led[i];
        for (int j = 0; j < n; j++) {
            // 消掉第 j 列的這項係數, 使 mat_{jc} = 0

```

```

    if (i == j) continue;
    int r = mat[j][c] / mat[i][c];
    // mat_j -= mat_i * (mat_{jc} / mat_{ic})
    for (int k = c; k <= m; k++)
        mat[j][k] = mat[j][k] - mat[i][k] * r;
}
for (int i = 0; i < m; i++) {
    // 紀錄 x_c 的解為 b_i / mat_{ic}
    ans[led[i]] = mat[i][m] / mat[i][led[i]];
}
return true; // 有解
}

```

3.13 約瑟夫問題

```

int josephus(int n, int m) { // n人每m次
    int ans = 0;
    for (int i = 1; i <= n; ++i)
        ans = (ans + m) % i;
    return ans;
}

```

4 Graph

4.1 Topological Sort

```

vector<int> edge[MAXN], ans;
int deg[MAXN]; // in degree
void topo(int n) { // 1-base
    queue<int> que;
    for (int i = 1; i <= n; i++)
        if (!deg[i]) que.push(i);
    while (!que.empty()) {
        int u = que.front();
        que.pop();
        ans.push_back(u);
        // 結果存 ans
        for (int v : edge[u]) {
            deg[v]--;
            if (!deg[v]) que.push(v);
        }
    }
}

```

4.2 Tarjan's Algorithm for BCC

```

struct BccVertex {
    int n, nScc, step, dfn[MXN], low[MXN];
    vector<int> E[MXN], sccv[MXN];
    int top, stk[MXN];
    void init(int _n) {
        n = _n;
        nScc = step = 0;
        for (int i = 0; i < n; i++)
            E[i].clear();
    }
    void addEdge(int u, int v) {
        E[u].PB(v);
        E[v].PB(u);
    }
    void DFS(int u, int f) {
        dfn[u] = low[u] = step++;
        stk[top++] = u;
        for (auto v : E[u]) {
            if (v == f) continue;
            if (dfn[v] == -1) {
                DFS(v, u);
                low[u] = min(low[u], low[v]);
                if (low[v] >= dfn[u]) {
                    int z;
                    sccv[nScc].clear();
                    do {
                        z = stk[--top];
                        sccv[nScc].PB(z);
                    } while (z != v);
                    sccv[nScc++].PB(u);
                }
            } else low[u] = min(low[u], dfn[v]);
        }
    }
    vector<vector<int>> solve() {
        vector<vector<int>> res;
        for (int i = 0; i < n; i++)

```

```

        dfn[i] = low[i] = -1;
        for (int i = 0; i < n; i++)
            if (dfn[i] == -1) {
                top = 0;
                DFS(i, i);
            }
        REP(i, nScc) res.PB(sccv[i]);
        return res;
    }
} graph;

```

4.3 Bellmen-Ford Algorithm

```

struct Edge {
    int from, to, weight;
} edge[MAXN];
int dis[MAXN];
void bellmen_ford(int n, int m) {
    dis[1] = 0; // 起點的距離一定是 0
    for (int j = 0; j < n - 1; j++) { // n 個節點要跑 n - 1 次
        for (int i = 0; i < m; i++) { // 對於所有邊都嘗試鬆弛
            if (dis[edge[i].to] >
                dis[edge[i].from] + edge[i].weight)
                dis[edge[i].to] =
                    dis[edge[i].from] + edge[i].weight;
        }
    }
}

```

4.4 Dijkstra's Algorithm

```

vector<pii> vec[N]; // vec[u] = {v, w}: u 為起點 v 為終點
// w 為路權
bool vis[N]; // 紀錄該節點是否走過
vector<int> dijkstra(int s) { // 起點
    vector<int> dis(N);
    for (int i = 0; i < N; i++) // 初始化
        dis[i] = INF; // 值要設為比可能的最短路徑權重要大的值
    dis[s] = 0;
    priority_queue<pii, vector<pii>, greater<pii>>
        pq; // 以小到大批序
    // {w, v}: w 為路權 v 為節點
    pq.push({dis[s], s});
    while (pq.empty() == 0) {
        int u = pq.top().second;
        pq.pop();
        if (vis[u]) // 走過的不要再走
            continue;
        vis[u] = 1;
        for (auto i : vec[u]) {
            int v = i.first, w = i.second;
            if (dis[u] + w < dis[v]) { // 鬆弛
                dis[v] = dis[u] + w;
                pq.push({dis[v], v});
            }
        }
    }
    return dis;
}

```

4.5 Dinic's Algorithm for Maximum Flow

```

#define SZ(x) ((int)x.size())
#define PB push_back
struct Dinic {
    struct Edge {
        int v, f, re;
    };
    int n, s, t, level[MXN];
    vector<Edge> E[MXN];
    void init(int _n, int _s, int _t) {
        n = _n;
        s = _s;
        t = _t;
        for (int i = 0; i < n; i++) E[i].clear();
    }
    void add_edge(int u, int v, int f) {
        E[u].PB({v, f, SZ(E[v])});
        E[v].PB({u, 0, SZ(E[u]) - 1});
    }
    bool BFS() {

```

```

    for (int i = 0; i < n; i++) level[i] = -1;
    queue<int> que;
    que.push(s);
    level[s] = 0;
    while (!que.empty()) {
        int u = que.front();
        que.pop();
        for (auto it : E[u]) {
            if (it.f > 0 && level[it.v] == -1) {
                level[it.v] = level[u] + 1;
                que.push(it.v);
            }
        }
    }
    return level[t] != -1;
}

int DFS(int u, int nf) {
    if (u == t) return nf;
    int res = 0;
    for (auto &it : E[u]) {
        if (it.f > 0 && level[it.v] == level[u] + 1) {
            int tf = DFS(it.v, min(nf, it.f));
            res += tf;
            nf -= tf;
            it.f -= tf;
            E[it.v][it.re].f += tf;
            if (nf == 0) return res;
        }
    }
    if (!res) level[u] = -1;
    return res;
}

int flow(int res = 0) {
    while (BFS()) res += DFS(s, 2147483647);
    return res;
}

} flow;
/*
    查找二分圖配對點：
    在 add_edge 時寫上 index[u].push_back({v,
    flow.E[u].size()});

    在 flow 完之後寫上這個
    for (int i = 1; i <= n; i++) { // 遍歷所有節點
        for (pair<int, int> j : index[i]) {
            if (flow.E[i][j].second - 1).f == 0)
                cout << i << ' ' << j.first << '\n';
        }
    }
*/

```

4.6 Dominator Tree

```

#define REP(i, s, e) for (int i = (s); i <= (e); i++)
#define REPD(i, s, e) for (int i = (s); i >= (e); i--)
struct DominatorTree { // O(N)
    int n, s;
    vector<int> g[MAXN], pred[MAXN];
    vector<int> cov[MAXN];
    int dfn[MAXN], nfd[MAXN], ts;
    int par[MAXN]; // idom[u] s到u的最後一個必經點
    int sdom[MAXN], idom[MAXN];
    int mom[MAXN], mn[MAXN];
    inline bool cmp(int u, int v) {
        return dfn[u] < dfn[v];
    }
    int eval(int u) {
        if (mom[u] == u) return u;
        int res = eval(mom[u]);
        if (cmp(sdom[mn[mom[u]]], sdom[mn[u]]))
            mn[u] = mn[mom[u]];
        return mom[u] = res;
    }
    void init(int _n, int _s) {
        ts = 0;
        n = _n;
        s = _s;
        REP(i, 1, n) g[i].clear(), pred[i].clear();
    }
    void addEdge(int u, int v) {
        g[u].push_back(v);
        pred[v].push_back(u);
    }
}

```

```

void dfs(int u) {
    ts++;
    dfn[u] = ts;
    nfd[ts] = u;
    for (int v : g[u])
        if (dfn[v] == 0) {
            par[v] = u;
            dfs(v);
        }
}

void build() {
    REP(i, 1, n) {
        dfn[i] = nfd[i] = 0;
        cov[i].clear();
        mom[i] = mn[i] = sdom[i] = i;
    }
    dfs(s);
    REPD(i, n, 2) {
        int u = nfd[i];
        if (u == 0) continue;
        for (int v : pred[u])
            if (dfn[v]) {
                eval(v);
                if (cmp(sdom[mn[v]], sdom[u]))
                    sdom[u] = sdom[mn[v]];
            }
        cov[sdom[u]].push_back(u);
        mom[u] = par[u];
        for (int w : cov[par[u]]) {
            eval(w);
            if (cmp(sdom[mn[w]], par[u])) idom[w] = mn[w];
            else idom[w] = par[u];
        }
        cov[par[u]].clear();
    }
    REP(i, 2, n) {
        int u = nfd[i];
        if (u == 0) continue;
        if (idom[u] != sdom[u]) idom[u] = idom[idom[u]];
    }
} domT;

```

4.7 Floyd-Warshall Algorithm

```

int dis[N][N];
void init(int n) {
    for (int i = 0; i <= n; i++) { // 初始化
        for (int j = 0; j <= n; j++)
            dis[i][j] = INF;
        dis[i][i] = 0;
    }
}

void floyd_warshall(int n) {
    for (int k = 0; k < n; k++) { // 窮舉中繼點 k
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) { // 窮舉點對 (i, j)
                dis[i][j] = min(dis[i][j], dis[i][k] + dis[k][j]);
            }
        }
    }
}

```

4.8 Maximum Clique

```

#define N 111 // 80 ~ 100 左右
struct MaxClique { // 0-base
    typedef bitset<N> Int;
    Int linkto[N], v[N];
    int n;
    void init(int _n) {
        n = _n;
        for (int i = 0; i < n; i++) {
            linkto[i].reset();
            v[i].reset();
        }
    }
    void addEdge(int a, int b) { v[a][b] = v[b][a] = 1; }
    int popcount(const Int& val) { return val.count(); }
    int lowbit(const Int& val) { return val._Find_first(); }
    int ans, stk[N];
    int id[N], di[N], deg[N];
    Int cans;
    void maxclique(int elem_num, Int candi) {

```

```

    if (elem_num > ans) {
        ans = elem_num;
        cans.reset();
        for (int i = 0; i < elem_num; i++)
            cans[id[stk[i]]] = 1;
    }
    int potential = elem_num + popcount(candi);
    if (potential <= ans) return;
    int pivot = lowbit(candi);
    Int smaller_candi = candi & (~linkto[pivot]);
    while (smaller_candi.count() && potential > ans) {
        int next = lowbit(smaller_candi);
        candi[next] = !candi[next];
        smaller_candi[next] = !smaller_candi[next];
        potential--;
        if (next == pivot ||
            (smaller_candi & linkto[next]).count()) {
            stk[elem_num] = next;
            maxclique(elem_num + 1, candi & linkto[next]);
        }
    }
}
int solve() {
    for (int i = 0; i < n; i++) {
        id[i] = i;
        deg[i] = v[i].count();
    }
    sort(id, id + n, [&](int id1, int id2) {
        return deg[id1] > deg[id2];
    });
    for (int i = 0; i < n; i++)
        di[id[i]] = i;
    for (int i = 0; i < n; i++)
        for (int j = 0; j < n; j++)
            if (v[i][j]) linkto[di[i]][di[j]] = 1;
    Int cand;
    cand.reset();
    for (int i = 0; i < n; i++)
        cand[i] = 1;
    ans = 1;
    cans.reset();
    cans[0] = 1;
    maxclique(0, cand);
    return ans;
}
} solver;

```

4.9 Kosaraju's Algorithm for SCC

```

struct Scc { // 新的 DAG 可能會有多重邊
#define PB push_back
    int n, nScc, vst[MXN], bln[MXN];
    vector<int> E[MXN], rE[MXN], vec, dag[MXN];
    set<int> vis[MXN];
    void init(int _n) {
        n = _n;
        for (int i = 0; i <= n; i++) {
            E[i].clear(), rE[i].clear();
            dag[i].clear(), vis[i].clear();
            bln[i] = -1;
        }
    }
    void addEdge(int u, int v) {
        E[u].PB(v);
        rE[v].PB(u);
    }
    void DFS(int u) {
        vst[u] = 1;
        for (auto v : E[u])
            if (!vst[v]) DFS(v);
        vec.PB(u);
    }
    void rDFS(int u) {
        vst[u] = 1;
        bln[u] = nScc; // 編號是 0-base
        for (auto v : rE[u])
            if (!vst[v]) rDFS(v);
    }
    void build_new_dag() {
        for (int i = 0; i < n; i++) {
            for (int &j : E[i]) {
                if (bln[i] != bln[j] &&
                    vis[bln[i]].count(bln[j]) == 0) {
                    dag[bln[i]].push_back(bln[j]);
                }
            }
        }
    }
}

```

```

        vis[bln[i]].insert(bln[j]);
    }
}
}
}
}
void solve() {
    nScc = 0;
    vec.clear();
    fill(vst, vst + n + 1, 0);
    for (int i = 0; i < n; i++)
        if (!vst[i]) DFS(i);
    reverse(vec.begin(), vec.end());
    fill(vst, vst + n + 1, 0);
    for (auto v : vec) {
        if (!vst[v]) {
            rDFS(v);
            nScc++;
        }
    }
    build_new_dag();
}
};

```

4.10 Kruskal's Algorithm for MST

```

void kruskal() {
    sort(graph, graph + m); // 將邊照大小排序
    int ans = 0;

    for (int i = 0; i < m; i++) { //>:)
        if (find(graph[i].u) != find(graph[i].v)) {
            // 如果兩點未聯通
            merge(graph[i].u, graph[i].v);
            // 將兩點設成同一個集合
            ans += graph[i].w; // 權重加進答案
            if (sz[find(graph[i].u)] == n)
                break; // 當並查集大小等價於樹內點的數量
        }
    }
}

```

4.11 Kuhn-Munkre Algorithm

```

struct KM { // max weight, for min negate the weights
    int n, mx[MXN], my[MXN], pa[MXN];
    ll g[MXN][MXN], lx[MXN], ly[MXN], sy[MXN];
    bool vx[MXN], vy[MXN];
    void init(int _n) { // 1-based
        n = _n; // 初始化
        for (int i = 1; i <= n; i++)
            fill(g[i], g[i] + n + 1, 0);
    }
    void addEdge(int x, int y, ll w) { // 加邊
        g[x][y] = w;
    }
    void augment(int y) {
        for (int x, z; y; y = z)
            x = pa[y], z = mx[x], my[y] = x, mx[x] = y;
    }
    void bfs(int st) {
        for (int i = 1; i <= n; ++i)
            sy[i] = INF, vx[i] = vy[i] = 0;
        queue<int> q;
        q.push(st);
        for (;;) {
            while (q.size()) {
                int x = q.front();
                q.pop();
                vx[x] = 1;
                for (int y = 1; y <= n; ++y)
                    if (!vy[y]) {
                        ll t = lx[x] + ly[y] - g[x][y];
                        if (t == 0) {
                            pa[y] = x;
                            if (!my[y]) {
                                augment(y);
                                return;
                            }
                        }
                        vy[y] = 1, q.push(my[y]);
                    } else if (sy[y] > t) pa[y] = x, sy[y] = t;
            }
            ll cut = INF;
        }
    }
}

```

```

    for (int y = 1; y <= n; ++y)
        if (!vy[y] && cut > sy[y]) cut = sy[y];
    for (int j = 1; j <= n; ++j) {
        if (vx[j]) lx[j] -= cut;
        if (vy[j]) ly[j] += cut;
        else sy[j] -= cut;
    }
    for (int y = 1; y <= n; ++y)
        if (!vy[y] && sy[y] == 0) {
            if (!my[y]) {
                augment(y);
                return;
            }
            vy[y] = 1, q.push(my[y]);
        }
    }
}

ll solve() { // 解決
    fill(mx, mx + n + 1, 0);
    fill(my, my + n + 1, 0);
    fill(ly, ly + n + 1, 0);
    fill(lx, lx + n + 1, -INF);
    for (int x = 1; x <= n; ++x)
        for (int y = 1; y <= n; ++y)
            lx[x] = max(lx[x], g[x][y]);
    for (int x = 1; x <= n; ++x)
        bfs(x);
    ll ans = 0;
    for (int y = 1; y <= n; ++y)
        ans += g[my[y]][y]; // 答案存在 my[]
    return ans;
}

} graph; // 找二分圖最大權完美匹配

```

4.12 Max-flow-min-cost

```

struct zkwflow {
    static const int maxN = 100000;
    struct Edge {
        int v, f, re;
        ll w;
    };
};
int n, s, t, ptr[maxN];
bool vis[maxN];
ll dis[maxN];
vector<Edge> E[maxN];
void init(int _n, int _s, int _t) {
    n = _n, s = _s, t = _t;
    for (int i = 0; i < n; i++)
        E[i].clear();
}
void add_edge(int u, int v, int f, ll w) {
    E[u].push_back({v, f, (int)E[v].size(), w});
    E[v].push_back({u, 0, (int)E[u].size() - 1, -w});
}
bool SPFA() {
    fill_n(dis, n, LLONG_MAX);
    fill_n(vis, n, false);
    queue<int> q;
    q.push(s);
    dis[s] = 0;
    while (!q.empty()) {
        int u = q.front();
        q.pop();
        vis[u] = false;
        for (auto &it : E[u]) {
            if (it.f > 0 && dis[it.v] > dis[u] + it.w) {
                dis[it.v] = dis[u] + it.w;
                if (!vis[it.v]) {
                    vis[it.v] = true;
                    q.push(it.v);
                }
            }
        }
    }
    return dis[t] != LLONG_MAX;
}
int DFS(int u, int nf) {
    if (u == t) return nf;
    int res = 0;
    vis[u] = true;
    for (int &i = ptr[u]; i < (int)E[u].size(); i++) {
        auto &it = E[u][i];
        if (it.f > 0 && dis[it.v] == dis[u] + it.w &&

```

```

        !vis[it.v]) {
            int tf = DFS(it.v, min(nf, it.f));
            res += tf, nf -= tf, it.f -= tf;
            E[it.v][it.re].f += tf;
            if (nf == 0) {
                vis[u] = false;
                break;
            }
        }
    }
    return res;
}
pair<int, ll> flow() {
    int flow = 0;
    ll cost = 0;
    while (SPFA()) {
        fill_n(ptr, n, 0);
        int f = DFS(s, INT_MAX);
        flow += f;
        cost += dis[t] * f;
    }
    return {flow, cost};
} // reset: do nothing
} flow;

```

4.13 Hungarian Algorithm for Maching

```

vector<int> g[MXN];
bool vis[MXN];
int S[MXN]; // S 為紀錄這個點與誰匹配
int n, ans; // n: 左集合數量, ans: 紀錄答案

bool dfs(int u) { // 找最大匹配
    for (int &v : g[u]) {
        if (!vis[v]) {
            vis[v] = true;
            if (S[v] == -1 || dfs(S[v])) {
                S[v] = u;
                return true;
            }
        }
    }
    return false;
}

void solve() { // 記得每次使用需清空 vis
    memset(S, -1, sizeof(S));
    ans = 0;
    for (int i = 1; i <= n; i++) { // 跑左集合
        memset(vis, 0, sizeof(vis));
        if (dfs(i)) ans++;
    }
}

```

4.14 Hungarian Algorithm for Weighted Maching

```

struct KM { // max weight, for min negate the weights
    int n, mx[MXN], my[MXN], pa[MXN];
    ll g[MXN][MXN], lx[MXN], ly[MXN], sy[MXN];
    bool vx[MXN], vy[MXN];
    void init(int _n) { // 1-based
        n = _n; // 初始化
        for (int i = 1; i <= n; i++)
            fill(g[i], g[i] + n + 1, 0);
    }
    void addEdge(int x, int y, ll w) { // 加邊
        g[x][y] = w;
    }
    void augment(int y) {
        for (int x, z; y; y = z)
            x = pa[y], z = mx[x], my[y] = x, mx[x] = y;
    }
    void bfs(int st) {
        for (int i = 1; i <= n; ++i)
            sy[i] = INF, vx[i] = vy[i] = 0;
        queue<int> q;
        q.push(st);
        for (;;) {
            while (q.size()) {
                int x = q.front();
                q.pop();
                vx[x] = 1;

```



```

    for (int y = 1; y <= n; ++y)
        if (!vy[y]) {
            ll t = lx[x] + ly[y] - g[x][y];
            if (t == 0) {
                pa[y] = x;
                if (!my[y]) {
                    augment(y);
                    return;
                }
                vy[y] = 1, q.push(my[y]);
            } else if (sy[y] > t) pa[y] = x, sy[y] = t;
        }
    ll cut = INF;
    for (int y = 1; y <= n; ++y)
        if (!vy[y] && cut > sy[y]) cut = sy[y];
    for (int j = 1; j <= n; ++j) {
        if (vx[j]) lx[j] -= cut;
        if (vy[j]) ly[j] += cut;
        else sy[j] -= cut;
    }
    for (int y = 1; y <= n; ++y)
        if (!vy[y] && sy[y] == 0) {
            if (!my[y]) {
                augment(y);
                return;
            }
            vy[y] = 1, q.push(my[y]);
        }
}
}
ll solve() { // 解決
    fill(mx, mx + n + 1, 0);
    fill(my, my + n + 1, 0);
    fill(ly, ly + n + 1, 0);
    fill(lx, lx + n + 1, -INF);
    for (int x = 1; x <= n; ++x)
        for (int y = 1; y <= n; ++y)
            lx[x] = max(lx[x], g[x][y]);
    for (int x = 1; x <= n; ++x)
        bfs(x);
    ll ans = 0;
    for (int y = 1; y <= n; ++y)
        ans += g[my[y]][y]; // 答案存在 my[]
    return ans;
}
} graph; // 找二分圖最大權完美匹配

```

5 Tree Algorithms

5.1 時間戳記 & 倍增表 (初始化)

```

const int LG_MXN = __lg(MXN) + 2;
vector<int> edge[MXN]; // i 有哪些小孩
int timing = 1, lgN, n;
int in[MXN], out[MXN], anc[MXN][LG_MXN], depth[MXN];
void dfs_init(int u, int f) { // 0 倍祖先、時間戳記
    in[u] = timing++; // 這時進入 u
    anc[u][0] = f; // now 的 0 倍祖先是 f
    for (int v : edge[u]) {
        if (v == f) continue;
        depth[v] = depth[u] + 1;
        dfs_init(v, u);
    }
    out[u] = timing++; // 這時離開 u
}
void build_table(int n) { // 初始化
    lgN = __lg(n) + 1;
    dfs_init(1, 0);
    for (int i = 1; i <= lgN; i++) // 建立倍增表
        for (int now = 1; now <= n; now++)
            anc[now][i] = anc[anc[now][i - 1]][i - 1];
}

```

5.2 LCA

```

bool is_ancestor(int u, int v) { // 判斷 u 是否 v 是祖先
    return (in[u] < in[v] && out[v] < out[u]);
}
int getLCA(int x, int y) { // 找最近共同祖先
    if (is_ancestor(x, y)) // 如果 x 為 y 的祖先
        return x; // 則 LCA 為 y
}

```

```

if (is_ancestor(y, x)) // 如果 y 為 x 的祖先
    return y; // LCA 為 y
for (int i = lgN; i >= 0; i--) // 逼近 LCA
    if (!is_ancestor(anc[x][i], y) && anc[x][i])
        x = anc[x][i];
return anc[x][0]; // 回傳此點的父節點即為答案
}

```

5.3 Euler Tour for LCA

```

#include <bits/stdc++.h>
using namespace std;
const int N = 1e5 + 5;
vector<int> tour, edge[N];
int timing = 0;
int in[N], out[N];
void dfs(int u, int fa) {
    tour.push_back(u); // 進入的時間點
    in[u] = timing++;
    for (auto v : edge[u]) {
        if (v == fa) continue;
        dfs(v, u);
        tour.push_back(u); // 走完其中一個孩子 再走回自己
    }
    out[u] = timing++;
}
void build(vector<int>& tour) { // 把 tour 丟進線段樹初始化
    seg.build(tour); // 除了 tour, 還要建每個點的深度
}
int query(int x, int y) {
    if (is_ancster(x, y)) return x; // LCA(x,y) = x
    if (is_ancster(y, x)) return y; // LCA(x,y) = y;
    // 如果「y的區間」在「x的區間」的左邊，就交換
    if (out[y] < in[x]) swap(x, y);
    // 取得區間內哪個位置的深度是最低的
    int index = seg.query(out[x], in[y]);
    return tour[index];
}

```

5.4 樹上差分

```

int tag[N], f[N], ans[N], root;
void add(int x, int y) {
    int lca = getLCA(x, y);
    tag[x]++;
    tag[y]++;
    tag[lca]--;
    if (lca != root) tag[f[lca]]--;
}
int dfs_add(int now, int fa) {
    int diff = tag[now];
    for (auto nxt : edge[now]) {
        if (nxt == fa) continue;
        diff += dfs_add(nxt, now);
    }
    return ans[now] = diff;
}

```

5.5 DSU on Tree

```

int find(int x) {
    if (f[x] == x) return x;
    else return f[x] = find(f[x]);
}
void merge(int x, int y) {
    x = find(x), y = find(y);
    if (sz[x] < sz[y]) swap(x, y);
    sz[x] += sz[y];
    f[y] = x;
}
void init() {
    // 初始化一開始大小都為1 (每群一開始都是獨立一個)
    for (int i = 1; i <= n; i++) sz[i] = 1;
    for (int i = 1; i <= n; i++) f[i] = i;
}

```

5.6 樹鏈剖分

```

const int MXN = 2e5 + 7;
int top[MXN], son[MXN], dfn[MXN], rnk[MXN], dep[MXN],
    father[MXN];
vector<int> edge[MXN];
int dfs1(int v, int fa, int d) {
}

```

```

int maxsz = -1, maxu, total = 1;
dep[v] = d;
father[v] = fa;
for (int u : edge[v]) {
    if (fa == u) continue;
    int temp = dfs1(u, v, d + 1);
    total += temp;
    if (temp > maxsz) {
        maxsz = temp;
        maxu = u;
    }
}
if (maxsz == -1) son[v] = -1;
else son[v] = maxu;
return total;
}
int times = 1;
void dfs2(int v, int fa) {
    rnk[times] = v;
    dfn[v] = times++;
    top[v] = (fa == -1 || son[fa] != v ? v : top[fa]);
    if (son[v] != -1) dfs2(son[v], v);
    for (int u : edge[v]) {
        if (fa == u || u == son[v]) continue;
        dfs2(u, v);
    }
}
// rnk: 剖分後的編號 (rnk[時間] = 原點)
// dfn: 剖分後的編號 (dfn[原點] = 時間)
// top: 剖分的頭頭 son: 剖分的重兒子

```

6 String Algorithms

6.1 Suffix Array

```

const int MXN = 100005;
struct SA{
    int n;
    vector<int> rk, oldrk, sa, h;
    string s;
    SA(string &t): s(t), n(t.size()), rk(n<<1), oldrk(n
        <<1), sa(n), h(n){
        for(int i = 0; i < n; i++) rk[i] = s[i] - 'a' + 1, sa[
            i] = i;
        for(int w = 1; w < n; w<=1){
            sort(sa.begin(), sa.end(), [&](int x, int y){
                return (rk[x]==rk[y]) ? (rk[x+w]<rk[y+w])
                    : (rk[x]<rk[y]);
            });
            oldrk = rk;
            rk[sa[0]] = 1;
            for(int i = 1, p = 1; i < n; i++){
                if(oldrk[sa[i]]==oldrk[sa[i-1]] && oldrk[
                    sa[i]+w]==oldrk[sa[i-1]+w]) rk[sa[i]]
                    = p;
                else rk[sa[i]] = ++p;
            }
        }
        for(int i = 0, k = 0; i < n; i++){
            if(rk[i] == 1) continue;
            if(k) k--;
            while(s[i+k] == s[sa[rk[i]-2]+k]) k++;
            h[rk[i]-1] = k;
        }
    }
    void find(string p){ // binary search
        int l=0, r=n, m=p.size();
        if(m > n) return;
        for(int i = 0; i < m; i++){
            auto cmp = [&](int lhs, int rhs){
                return s[lhs+i] < rhs;
            };
            auto nl = lower_bound(sa+l, sa+r, p[i], cmp) -
                sa;
            auto nr = lower_bound(sa+l, sa+r, p[i]+1, cmp)
                - sa;
            l = nl, r = nr;
            if(l >= r) break;
        }
        cout << (r > 1 ? "YES" : "NO") << "\n";
    }
};

```

6.2 Suffix Array (SAIS)

```

// 此為後綴陣列模板
// 用法:
// 把要處理的字串轉成整數陣列傳入suffix_array(), 會得到SA
// 與H維結果
const int N = 500010; // 有可能要在這就寫原本字串的兩倍長
struct SA {
#define REP(i, n) for (int i = 0; i < int(n); i++)
#define REPl(i, a, b) for (int i = (a); i <= int(b); i++)
    bool _t[N * 2];
    int _s[N * 2], _sa[N * 2], _c[N * 2], x[N], _p[N],
        _q[N * 2], hei[N], r[N];
    int operator[](int i) { return _sa[i]; }
    void build(int *s, int n, int m) {
        memcpy(_s, s, sizeof(int) * n);
        sais(_s, _sa, _p, _q, _t, _c, n, m);
        mkhei(n);
    }
    void mkhei(int n) {
        REP(i, n) r[_sa[i]] = i;
        hei[0] = 0;
        REP(i, n) if (r[i]) {
            int ans = i > 0 ? max(hei[r[i] - 1] - 1, 0) : 0;
            while (_s[i + ans] == _s[_sa[r[i] - 1] + ans])
                ans++;
            hei[r[i]] = ans;
        }
    }
    void sais(int *s, int *sa, int *p, int *q, bool *t,
        int *c, int n, int z) {
        bool uniq = t[n - 1] = true, neq;
        int nn = 0, nmzx = -1, *nsa = sa + n, *ns = s + n,
            lst = -1;
#define MS0(x, n) memset((x), 0, n * sizeof(*(x)))
#define MAGIC(XD) \
        MS0(sa, n); \
        memcpy(x, c, sizeof(int) * z); \
        XD; \
        memcpy(x + 1, c, sizeof(int) * (z - 1)); \
        REP(i, n) \
        if (sa[i] && !t[sa[i] - 1]) \
            sa[x[s[sa[i] - 1]]++] = sa[i] - 1; \
        memcpy(x, c, sizeof(int) * z); \
        for (int i = n - 1; i >= 0; i--) \
            if (sa[i] && t[sa[i] - 1]) \
                sa[--x[s[sa[i] - 1]]] = sa[i] - 1; \
        MS0(c, z); \
        REP(i, n) uniq &= ++c[s[i]] < 2; \
        REP(i, z - 1) c[i + 1] += c[i]; \
        if (uniq) { \
            REP(i, n) sa[--c[s[i]]] = i; \
            return; \
        } \
        for (int i = n - 2; i >= 0; i--) \
            t[i] = \
                (s[i] == s[i + 1] ? t[i + 1] : s[i] < s[i + 1]); \
        MAGIC(REPl(i, 1, n - 1) if (t[i] && !t[i - 1]) \
            sa[--x[s[i]]] = p[q[i] = nn++] = i); \
        REP(i, n) if (sa[i] && t[sa[i]] && !t[sa[i] - 1]) { \
            neq = lst < 0 || memcmp(s + sa[i], s + lst, \
                (p[q[sa[i]] + 1] - sa[i]) * \
                sizeof(int)); \
            ns[q[lst = sa[i]]] = nmzx += neq; \
        } \
        sais(ns, nsa, p + nn, q + n, t + n, c + z, nn, \
            nmzx + 1); \
        MAGIC(for (int i = nn - 1; i >= 0; i--) \
            sa[--x[s[p[nsa[i]]]]] = p[nsa[i]]); \
    }
    } sa;
    int H[N], SA[N]; // SA: 後綴陣列的length
        // H[i]: SA[i]與SA[i - 1]的共同子字串長
    void suffix_array(int *ip, int len) {
        // should padding a zero in the back
        // ip is int array, len is array length
        // ip[0..n-1] != 0, and ip[len] = 0
        ip[len++] = 0;
        sa.build(ip, len, 128);
        for (int i = 0; i < len; i++) {
            H[i] = sa.hei[i + 1];
            SA[i] = sa._sa[i + 1];
        }
    }
}

```

```
// resulting height, sa array \in [0,len)
}
```

6.3 Suffix Automata

```
struct SAM{
    struct State{
        int len;
        State *link;
        map<char, State*> next;
    };
    State *first, *last;
    SAM(){first = last = new State();}
    SAM(string &s): SAM(){
        for(int i = 0; i < s.size(); i++) add(s[i]);
    }
    void add(char ch){
        State* nw = new State();
        nw->len = last->len+1;
        nw->link = first;
        State *cur = last;
        while(cur && !cur->next[ch]){
            cur->next[ch] = nw;
            cur = cur->link;
        }
        if(cur && cur->next[ch] != nw){
            State *mid = cur->next[ch];
            if(cur->len+1 == mid->len){
                nw->link = mid;
            }
            else{
                State *clone = new State(*mid);
                clone->len = cur->len+1;
                nw->link = mid->link = clone;
                while(cur && cur->next[ch] == mid){
                    cur->next[ch] = clone;
                    cur = cur->link;
                }
            }
        }
        last = nw;
    }
};
```

6.4 Z-algorithm

```
int z[MAXN];
void Z_value(const string& s) {
    // z[i] = lcp(s[1...], s[i...])
    int i, j, left, right, len = s.size();
    left = right = 0;
    z[0] = len;
    for (i = 1; i < len; i++) {
        j = max(min(z[i - left], right - i), 0);
        for (; i + j < len && s[i + j] == s[j]; j++);
        z[i] = j;
        if (i + z[i] > right) {
            right = i + z[i];
            left = i;
        }
    }
}
```

6.5 Knuth-Morris-Pratt Algorithm

```
vector<int> KMP(string s, string t){
    s = t + '@' + s;
    int sz = s.size();
    vector<int> pi(sz);
    for(int i = 1; i < sz; i++){
        int len = pi[i-1];
        while(len != 0 && s[i] != s[len]) len = pi[len-1];
        if(s[i] == s[len]) pi[i] = len+1;
    }
    return pi;
}
```

6.6 Manacher's Algorithm

```
int ans[MAXN << 1];
char s[MAXN << 1];
string ret; // 迴文字串儲存結果
void find_palindrome(char *s, int len, int *z) {
```

```
int idx = 0, mx = -1;
for (int i = 0; i < len; i++) { // 找迴文半徑最大值
    if (mx < ans[i]) {
        idx = i;
        mx = ans[i];
    }
}
for (int i = idx - mx + 1; i < idx; i++)
    if (s[i] != '@') ret.push_back(s[i]);
for (int i = idx; i <= idx + mx - 1; i++)
    if (s[i] != '@') ret.push_back(s[i]);
// cout << ret << '\n'; //// 直接輸出
}
void z_value_pal(char *s, int len,
                 int *z) { // 字串，長度，輸出的陣列
    len = (len << 1) + 1;
    for (int i = len - 1; i >= 0; i--)
        s[i] = i & 1 ? s[i >> 1] : '@';
    z[0] = 1;
    for (int i = 1, l = 0, r = 0; i < len; i++) {
        z[i] = i < r ? min(z[l + 1 - i], r - i) : 1;
        while (i - z[i] >= 0 && i + z[i] < len &&
                s[i - z[i]] == s[i + z[i]])
            ++z[i];
        if (i + z[i] > r) l = i, r = i + z[i];
    }
    find_palindrome(s, strlen(s), z);
}
```

6.7 Minimum Rotation

```
int min_rotation(string s) {
    int a = 0, N = s.size();
    s += s;
    for (int i = 0; i < N; i++) {
        for (int j = 0; j < N; j++) {
            if (a + j == i || s[a + j] < s[i + j]) {
                i += max(0, j - 1);
                break;
            }
            if (s[a + j] > s[i + j]) {
                a = i;
                break;
            }
        }
    }
    return a;
}
// rotate(begin(s), begin(s)+minRotation(s), end(s))
// 上面註解可以直接讓一個字串做 rotate，得到字典序最小的
// rotation
```

6.8 Rolling Hash

```
ll P1 = 75577, P2 = 12721, MOD = 998244353;
pii hashes[MAXN], hashes2[MAXN];
string s1, s2;
void build(string s, pii hash) {
    ll val1 = 0, val2 = 0;
    for (int i = 0; i < s.length(); i++) {
        val1 = (val1 * P1 % MOD + s[i]) % MOD;
        val2 = (val2 * P2 % MOD + s[i]) % MOD;
        hash[i] = {val1, val2};
    }
}
ll minus(ll a, ll b) { return (a - b + MOD) % MOD; }
ll cnt_occur() { // 檢查
    int n = s1.size(), k = s2.size();
    ll cnt = (hashes2[k - 1].x == hashes[k - 1].x &&
              hashes2[k - 1].y == hashes[k - 1].y);
    ll val1 = 0, val2 = 0;
    for (int i = 1, j = k; j < n; j++, i++) {
        val1 = minus(hashes[j].x,
                      hashes[i - 1].x * fpow(P1, j - i + 1));
        val2 = minus(hashes[j].y,
                      hashes[i - 1].y * fpow(P2, j - i + 1));
        if (hashes2[k - 1].x == val1 &&
            hashes2[k - 1].y == val2)
            cnt++;
    }
    return cnt;
}
```

6.9 Another Hash

```
const int M = 1e9+7;
const int C = 13331;
struct HashString{
    int n;
    vector<ll> pows, sums;
    HashString(string s):n(s.size()),pows(n + 1, 1),sums(n + 1){
        for (int i = 1; i <= n; i++) {
            pows[i] = pows[i - 1] * C % M;
            sums[i] = (sums[i - 1] * C + s[i - 1]) % M;
        }
    }
    ll hash(int l, int r) {
        ll h = sums[r] - sums[l] * pows[r - l];
        return (h % M + M) % M;
    }
};
```

7 Geometry

7.1 Basic Geometry Structure

```
int dcmp(ld x) { // 處理小數點精度的東東
    if (abs(x) < EPS) return 0;
    else return x < 0 ? -1 : 1;
}
struct Pt {
    ld x, y;
    Pt(ld _x = 0, ld _y = 0) : x(_x), y(_y) {}
};
Pt operator+(const Pt &a, const Pt &b) {
    return Pt(a.x + b.x, a.y + b.y);
}
Pt operator-(const Pt &a, const Pt &b) {
    return Pt(a.x - b.x, a.y - b.y);
}
Pt operator*(const Pt &a, const ld &b) {
    return Pt(a.x * b, a.y * b);
}
Pt operator/(const Pt &a, const ld &b) {
    return Pt(a.x / b, a.y / b);
}
ld operator*(const Pt &a, const Pt &b) { // 內積
    return a.x * b.x + a.y * b.y;
}
ld operator^(const Pt &a, const Pt &b) { // 行列式 (叉乘)
    return a.x * b.y - a.y * b.x;
}
bool operator<(const Pt &a, const Pt &b) {
    return a.x < b.x || (a.x == b.x && a.y < b.y);
    // return dcmp(a.x - b.x) < 0 ||
    // (dcmp(a.x - b.x) == 0 && dcmp(a.y - b.y) < 0);
}
ld cross(Pt a, Pt b, Pt o) { // 行列式(叉乘)，存入三個點
    Pt p1 = a - o, p2 = b - o;
    return p1.x * p2.y - p1.y * p2.x;
}
bool operator==(const Pt &a, const Pt &b) {
    return dcmp(a.x - b.x) == 0 && dcmp(a.y - b.y) == 0;
}
// 平方
ld norm2(const Pt &a) { return a * a; }
ld norm(const Pt &a) { return sqrt(norm2(a)); }
// 與原本向量垂直的向量
Pt perp(const Pt &a) { return Pt(-a.y, a.x); }
// 旋轉矩陣
Pt rotate(const Pt &a, ld rad) {
    return Pt(a.x * cos(rad) - a.y * sin(rad),
              a.x * sin(rad) + a.y * cos(rad));
}
struct Line {
    Pt s, e, v; // start, end, end-start
    ld ang; // 斜率的樣子?
    Line(Pt _s = Pt(0, 0), Pt _e = Pt(0, 0))
        : s(_s), e(_e) {
        v = e - s;
        ang = atan2(v.y, v.x);
    }
    bool operator<(const Line &L) const {
```

```
        return ang < L.ang;
    }
};
struct Circle {
    Pt o;
    ld r;
    Circle(Pt _o = Pt(0, 0), ld _r = 0) : o(_o), r(_r) {}
};
```

7.2 Circle Cover Area

```
struct CircleCover {
    int C;
    Circle c[N]; // 填入C(圓數量), c(圓陣列)
    bool g[N][N], overlap[N][N];
    // Area[i] : area covered by at least i circles
    D Area[N];
    void init(int _C) { C = _C; }
    bool CCinter(Circle &a, Circle &b, Pt &p1, Pt &p2) {
        Pt o1 = a.o, o2 = b.o;
        D r1 = a.r, r2 = b.r;
        if (norm(o1 - o2) > r1 + r2) return {};
        if (norm(o1 - o2) < max(r1, r2) - min(r1, r2))
            return {};
        D d2 = (o1 - o2) * (o1 - o2);
        D d = sqrt(d2);
        if (d > r1 + r2) return false;
        Pt u = (o1 + o2) * 0.5 +
              (o1 - o2) * ((r2 * r2 - r1 * r1) / (2 * d2));
        D A = sqrt((r1 + r2 + d) * (r1 - r2 + d) *
                  (r1 + r2 - d) * (-r1 + r2 + d));
        Pt v = Pt(o1.y - o2.y, -o1.x + o2.x) * A / (2 * d2);
        p1 = u + v;
        p2 = u - v;
        return true;
    }
    struct Teve {
        Pt p;
        D ang;
        int add;
        Teve() {}
        Teve(Pt _a, D _b, int _c) : p(_a), ang(_b), add(_c) {}
        bool operator<(const Teve &a) const {
            return ang < a.ang;
        }
    } eve[N * 2];
    // strict: x = 0, otherwise x = -1
    bool disjunct(Circle &a, Circle &b, int x) {
        return dcmp(norm(a.o - b.o) - a.r - b.r) > x;
    }
    bool contain(Circle &a, Circle &b, int x) {
        return dcmp(a.r - b.r - norm(a.o - b.o)) > x;
    }
    bool contain(int i, int j) {
        /* c[j] is non-strictly in c[i]. */
        return (dcmp(c[i].r - c[j].r) > 0 ||
                (dcmp(c[i].r - c[j].r) == 0 && i < j)) &&
                contain(c[i], c[j], -1);
    }
    void solve() {
        for (int i = 0; i <= C + 1; i++) Area[i] = 0;
        for (int i = 0; i < C; i++)
            for (int j = 0; j < C; j++)
                overlap[i][j] = contain(i, j);
        for (int i = 0; i < C; i++)
            for (int j = 0; j < C; j++)
                g[i][j] = !(overlap[i][j] || overlap[j][i] ||
                            disjunct(c[i], c[j], -1));
        for (int i = 0; i < C; i++) {
            int E = 0, cnt = 1;
            for (int j = 0; j < C; j++)
                if (j != i && overlap[j][i]) cnt++;
            for (int j = 0; j < C; j++)
                if (i != j && g[i][j]) {
                    Pt aa, bb;
                    CCinter(c[i], c[j], aa, bb);
                    D A = atan2(aa.y - c[i].o.y, aa.x - c[i].o.x);
                    D B = atan2(bb.y - c[i].o.y, bb.x - c[i].o.x);
                    eve[E++] = Teve(bb, B, 1);
                    eve[E++] = Teve(aa, A, -1);
                    if (B > A) cnt++;
                }
            if (E == 0) Area[cnt] += pi * c[i].r * c[i].r;
```

```

else {
    sort(eve, eve + E);
    eve[E] = eve[0];
    for (int j = 0; j < E; j++) {
        cnt += eve[j].add;
        Area[cnt] += (eve[j].p ^ eve[j + 1].p) * 0.5;
        D theta = eve[j + 1].ang - eve[j].ang;
        if (theta < 0) theta += 2.0 * pi;
        Area[cnt] += (theta - sin(theta)) * c[i].r *
            c[i].r * 0.5;
    }
}
}
} ret;

```

7.3 圓的公切線

```

vector<Line> go(const Circle& c1, const Circle& c2,
    int sign1) {
    // sign1 = 1 for outer tang, -1 for inner tang
    vector<Line> ret;
    double d_sq = norm2(c1.o - c2.o);
    if (d_sq < EPS) return ret;
    double d = sqrt(d_sq);
    Pt v = (c2.o - c1.o) / d;
    double c = (c1.r - sign1 * c2.r) / d;
    if (c * c > 1) return ret;
    double h = sqrt(max(0.0, 1.0 - c * c));
    for (int sign2 = 1; sign2 >= -1; sign2 -= 2) {
        Pt n = {v.x * c - sign2 * h * v.y,
            v.y * c + sign2 * h * v.x};
        Pt p1 = c1.o + n * c1.r;
        Pt p2 = c2.o + n * (c2.r * sign1);
        if (fabs(p1.x - p2.x) < EPS and
            fabs(p1.y - p2.y) < EPS)
            p2 = p1 + perp(c2.o - c1.o);
        ret.push_back({p1, p2});
    }
    return ret;
}

```

7.4 圓的交點

```

vector<Pt> interCircle(Pt o1, ld r1, Pt o2, ld r2) {
    if (norm(o1 - o2) > r1 + r2) return {};
    if (norm(o1 - o2) < max(r1, r2) - min(r1, r2))
        return {};
    ld d2 = (o1 - o2) * (o1 - o2);
    ld d = sqrt(d2);
    if (d > r1 + r2) return {};
    Pt u = (o1 + o2) * 0.5 +
        (o1 - o2) * ((r2 * r2 - r1 * r1) / (2 * d2));
    ld A = sqrt((r1 + r2 + d) * (r1 - r2 + d) *
        (r1 + r2 - d) * (-r1 + r2 + d));
    Pt v = Pt(o1.y - o2.y, -o1.x + o2.x) * A / (2 * d2);
    return {u + v, u - v};
}

```

7.5 Convex Hull

```

double cross(Pt o, Pt a, Pt b) {
    return (a - o) ^ (b - o);
}
vector<Pt> convex_hull(vector<Pt> pt) {
    sort(pt.begin(), pt.end());
    int top = 0;
    vector<Pt> stk(2 * pt.size());
    for (int i = 0; i < (int)pt.size(); i++) {
        while (top >= 2 &&
            cross(stk[top - 2], stk[top - 1], pt[i]) <= 0)
            top--;
        stk[top++] = pt[i];
    }
    for (int i = pt.size() - 2, t = top + 1; i >= 0; i--) {
        while (top >= t &&
            cross(stk[top - 2], stk[top - 1], pt[i]) <= 0)
            top--;
        stk[top++] = pt[i];
    }
    stk.resize(top - 1);
    return stk;
}

```

7.6 Convex Hull Tricks

```

struct Convex {
    int n;
    vector<Pt> A, V, L, U;
    Convex(const vector<Pt> &_A)
        : A(_A), n(_A.size()) { // n >= 3
        rotate(_A.begin(), min_element(all(_A)), _A.end());
        // 最左的點排最前面 有時可以不需要這一行
        auto it = max_element(all(A));
        L.assign(A.begin(), it + 1);
        U.assign(it, A.end()), U.push_back(A[0]);
        for (int i = 0; i < n; i++) {
            V.push_back(A[(i + 1) % n] - A[i]);
        }
    }
    int PtSide(Pt p, Line L) {
        return dcmp((L.b - L.a) ^ (p - L.a));
    }
    int inside(Pt p, const vector<Pt> &h, auto f) {
        auto it = lower_bound(all(h), p, f);
        if (it == h.end()) return 0;
        if (it == h.begin()) return p == *it;
        return 1 - dcmp((p - *prev(it)) ^ (*it - *prev(it)));
    }
    // 1. whether a given point is inside the CH
    // ret 0: out, 1: on, 2: in
    int inside(Pt p) {
        return min(inside(p, L, less{}),
            inside(p, U, greater{}));
    }
    static bool cmp(Pt a, Pt b) { return dcmp(a ^ b) > 0; }
    // 2. Find tangent points of a given vector
    // ret the idx of far/closer tangent point
    int tangent(Pt v, bool close = true) {
        assert(v != Pt{});
        auto l = V.begin(), r = V.begin() + L.size() - 1;
        if (v < Pt{}) l = r, r = V.end();
        if (close)
            return (lower_bound(l, r, v, cmp) - V.begin()) % n;
        return (upper_bound(l, r, v, cmp) - V.begin()) % n;
    }
    // 3. Find 2 tang pts on CH of a given outside point
    // return index of tangent points
    // return {-1, -1} if inside CH
    array<int, 2> tangent2(Pt p) {
        array<int, 2> t{-1, -1};
        if (inside(p) == 2) return t;
        if (auto it = lower_bound(all(L), p);
            it != L.end() and p == *it) {
            int s = it - L.begin();
            return {(s + 1) % n, (s - 1 + n) % n};
        }
        if (auto it = lower_bound(all(U), p, greater{});
            it != U.end() and p == *it) {
            int s = it - U.begin() + L.size() - 1;
            return {(s + 1) % n, (s - 1 + n) % n};
        }
        for (int i = 0; i != t[0];
            i = tangent((A[t[0] = i] - p), 0));
        for (int i = 0; i != t[1];
            i = tangent((p - A[t[1] = i]), 1));
        return t;
    }
    int find(int l, int r, Line L) {
        if (r < l) r += n;
        int s = PtSide(A[l % n], L);
        return *ranges::partition_point(
            views::iota(l, r),
            [&](int m) {
                return PtSide(A[m % n], L) == s;
            }) -
            1;
    }
    // 4. Find intersection point of a given line
    // intersection is on edge (i, next(i))
    vector<int> intersect(Line L) {
        int l = tangent(L.a - L.b), r = tangent(L.b - L.a);
        if (PtSide(A[l], L) == 0) return {l};
        if (PtSide(A[r], L) == 0) return {r};
        if (PtSide(A[l], L) * PtSide(A[r], L) > 0) return {};
        return {find(l, r, L) % n, find(r, l, L) % n};
    }
};

```

7.7 Finding Farthest Point using CH

```
double FarthestPair(vector<Pt> arr) {
    double ret = 0; // 傳入凸包
    int sz = sz;
    for (int i = 0, j = i + 1; i < sz; i++) {
        while (distance(arr[i], arr[j]) <=
                distance(arr[i], arr[(j + 1) % sz])) {
            j = (j + 1) % sz;
        }
        ret = max(ret, distance(arr[i], arr[j]));
    }
    return ret;
}
```

7.8 Pick’s Theory

$$A = i + \frac{b}{2} - 1, \quad 2A = 2i + b - 2, \quad i = A - \frac{b}{2} + 1$$

其中 A 為面積、 b 為圖形內部的點數量、 i 為圖形線上的點數量

```
vector<Pt> p;
ll compute_area(int n) { // A * 2
    ll area = 0;
    for (int i = 1; i < n; i++)
        area += cross(p[i], p[(i + 1) % n], p[0]);
    area = abs(area);
    return area;
}
ll compute_pt_on_line(int n) { // b
    ll pt = 0;
    for (int i = 0; i < n; i++)
        pt += __gcd(abs(p[i].x - p[(i + 1) % n].x),
                    abs(p[i].y - p[(i + 1) % n].y));
    return pt; // 這是網路上找的
}
ll compute_inner_pt(int n) { // i
    return (compute_area(n) - compute_pt_on_line(n)) / 2 + 1;
}
```

8 DP

8.1 區間 DP

時間複雜度 $O(n^3)$ ，常見的 n 在 500 以內，至多 1,000。

```
// 區間大小為 1 為 base case
for (int i = 1; i <= n; i++) dp[i][i] = 0;
// 由小區間往大開始轉移
for (int len = 2; len <= n; len++)
    for (int l = 1, r = len; r <= n; l++, r++) {
        dp[l][r] = INF;
        for (int k = l; k < r; k++)
            dp[l][r] = min(dp[l][r], dp[l][k] + dp[k + 1][r] +
                            n[i] * m[k] * n[r]);
    }
return dp[1][n];
```

8.2 位數 DP

dp[位數][狀態]，dp[pos][state]:
定義為目前位數在前導狀態為 state 的時候的計數
例如：求數字沒有出現 66 的數量 1 ~ r
→ dp[pos][1] 可表示計算 pos 個位數在前導出現一個 6 的計數
→ dp[3][1] 則計算 6XXX
模板的 pos 是反過來的，但不影響 (只是用來 dp 記憶用)

pos: 目前位數
state: 前導狀態
lead: 是否有前導 0
(大部分題目不用但有些數字 e.g. 00146 如果有影響時要考慮)
limit: 是否窮舉有被 num 限制

```
vector<int> num;
int dp[20][state];
int dfs(int pos, int state, bool lead, bool limit) {
    if (pos == num.size()) { // 有時要根據不同state回傳情況
        return 1;
    }
    if (limit == false && lead == false &&
        dp[pos][state] != -1)
        return dp[pos][state];
    int up = limit ? num[pos] : 9;
    int ans = 0;
    for (int i = 0; i <= up; i++) {
        // 有時要考慮那些狀況要continue
```

```
        ans += dfs(pos + 1, state || (check[i] == 2),
                    lead && i == 0, limit && i == num[pos]);
    }
    if (limit == false && lead == false)
        dp[pos][state] = ans;
    return ans;
}
```

8.3 SOS DP

```
for (int mask = 0; mask < (1 << N); mask++)
    F[mask] = A[mask];
for (int i = 0; i < N; i++)
    for (int mask = 0; mask < (1 << N); mask++)
        if (mask & (1 << i)) F[mask] += F[mask ^ (1 << i)];
```

9 Sorting

9.1 Merge Sort

```
template <typename T>
void merge_sort(T &arr, T &tmp, int l, int r) {
    // 陣列元素少於 1 直接結束
    if (r <= l + 1) return;

    // 分成左右兩邊處理
    int m = l + (r - l) / 2;
    merge_sort(arr, tmp, l, m); // [l, m)
    merge_sort(arr, tmp, m, r); // [m, r)

    // 合併兩邊的結果
    for (int i = l, li = l, ri = m; i < r; i++) {
        // 只剩下右邊 // 兩邊都有剩，右邊比左邊小
        if (li == m || (ri != r && arr[ri] < arr[li]))
            tmp[i] = arr[ri++];
        // 只剩下左邊 // 兩邊都有剩，左邊比右邊小
        else tmp[i] = arr[li++];
    }

    // 把結果複製回來
    for (int i = l; i < r; i++) arr[i] = tmp[i];
}
```

9.2 Quick Sort

```
template <typename T>
void quick_sort(T arr[], int l, int r) {
    // 陣列元素少於 1 直接結束
    if (r <= l + 1)
        return;

    // 左邊和右邊的指針
    int li = l + 1, ri = r - 1;
    // 判定的基準點
    T pivot = arr[l];

    // TODO: 左邊 <= 基準點 < 右邊
    while (li != ri) {
        // 右邊的指針往左移動，找比基準點小的值
        while (arr[ri] > pivot && li < ri)
            ri--;
        // 左邊的指針往右移動，找比基準點大的值
        while (arr[li] <= pivot && li < ri)
            li++;
        // 當左右指針沒有相遇時，表示不滿足 TODO
        if (li < ri)
            swap(arr[li], arr[ri]);
    }

    // 將基準點移到指針相遇的地方
    arr[l] = arr[li];
    arr[li] = pivot;

    quick_sort(arr, l, li); // 處理基準點的左邊
    quick_sort(arr, li + 1, r); // 處理基準點的右邊
}
```

10 Miscellaneous

10.1 GNU Builtins

A black and white photograph of a person in a dark, hooded costume with a long, flowing skirt, riding a broomstick against a dark background with small white specks.

10.7 Graph Paper





